

PrintGuard

The Safety Layer for Additive Manufacturing

Local screening. Auditable decisions. Practical compliance for the next era of 3D printing.

PrintGuard is a defensive classifier and review system for 3D-printing files. It helps organizations identify firearm-related risk before a file is published, accepted into a print queue, or fabricated in a managed environment.

96.75%

0.06%

<1 second

10,000 files

FIREARM DETECTION

FALSE-POSITIVE RATE

PER-FILE SCORING

SOURCE-SEPARATED TEST SET

WHY NOW

3D printers are becoming ordinary infrastructure for schools, labs, manufacturers, marketplaces, makerspaces, and homes. The same accessibility that makes additive manufacturing valuable also creates a public-safety gap: unserialized firearms, regulated components, and illegal accessories can be produced from digital files outside traditional compliance checkpoints.

Regulators are moving quickly. **California AB 2047** — the Firearm Printing Prevention Act — would require every 3D printer sold in the state to include firearm blocking technology, and it sets explicit performance standards for the detection algorithm: false-positive and false-negative thresholds, benchmark datasets, regular updates, and possible nongovernmental standards. New York, Washington, Colorado, and other states have moved on parallel tracks.

The market now needs a standard that actually works: accurate enough to reduce risk, transparent enough for public accountability, and careful enough to protect lawful 3D printing.

THE GAP

A credible standard cannot rely on filenames, keyword filters, or static hash lists alone. File names can be changed. Hashes break after small edits. Many lawful mechanical parts share geometry with restricted components. And broad surveillance would damage schools, small manufacturers, researchers, repair shops, and legitimate makers.

PrintGuard fills the gap with a local-first, evidence-backed review layer.

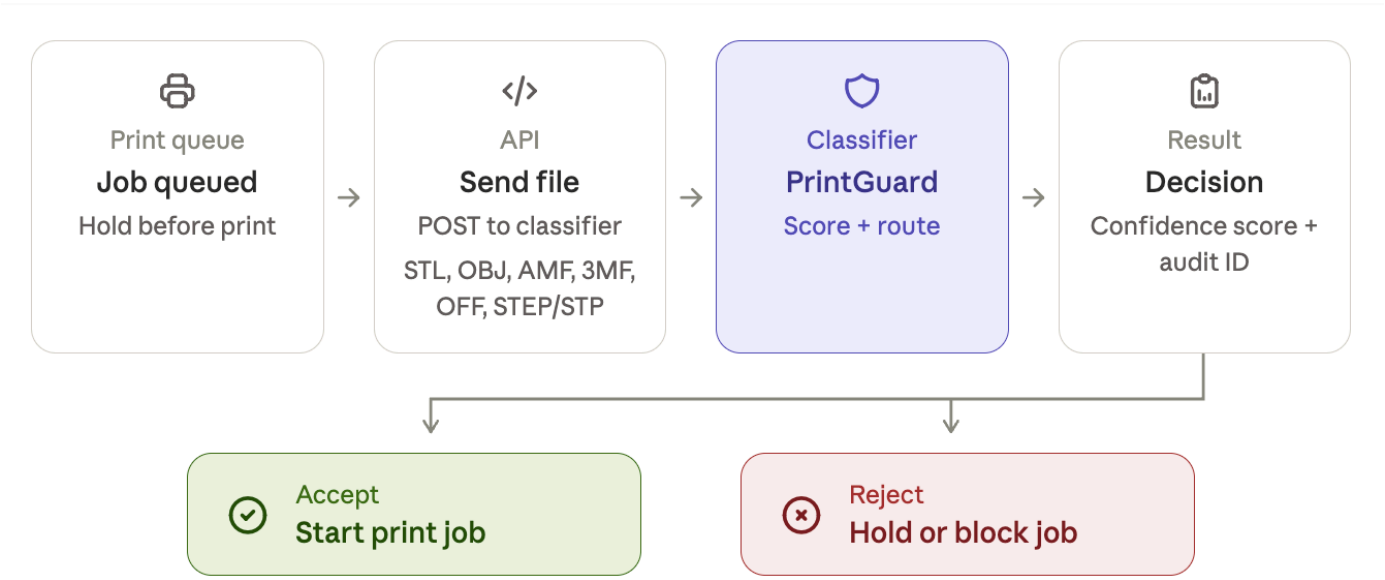
WHAT PRINTGUARD DOES

- Screens STL, OBJ, AMF, 3MF, and OFF mesh files.
- Routes STEP/STP CAD files for conversion before geometry scoring.
- Extracts shape-aware features from mesh statistics, point clouds, voxel occupancy, and multiview projections.
- Compares submissions against exact SHA-256 known-file evidence and near-geometry matches.
- Runs machine-learning and CNN classifiers on extracted parameters and mesh geometry.

- Returns a decision and confidence score for review.
- Logs hashes, model scores, evidence, and decisions for reproducible audit.
- Deploys as a standalone screening gate or as the first stage of a human review pipeline for higher-volume operations.

CLASSIFIER CONNECTION

Returns results within a second for average file sizes.



CURRENT V1 ASSETS

ASSET	CURRENT STATE
Binary training manifest	113,368 mesh rows
Hard-negative mechanical examples	99,848 rows
Restricted positive mesh examples	13,520 rows
Review surfaces	CLI, Python package, local web portal, JSON reports
Evidence layer	Exact hash match and near-geometry match
Model stack	Logistic baseline, nearest-centroid baseline, optional multiview CNN

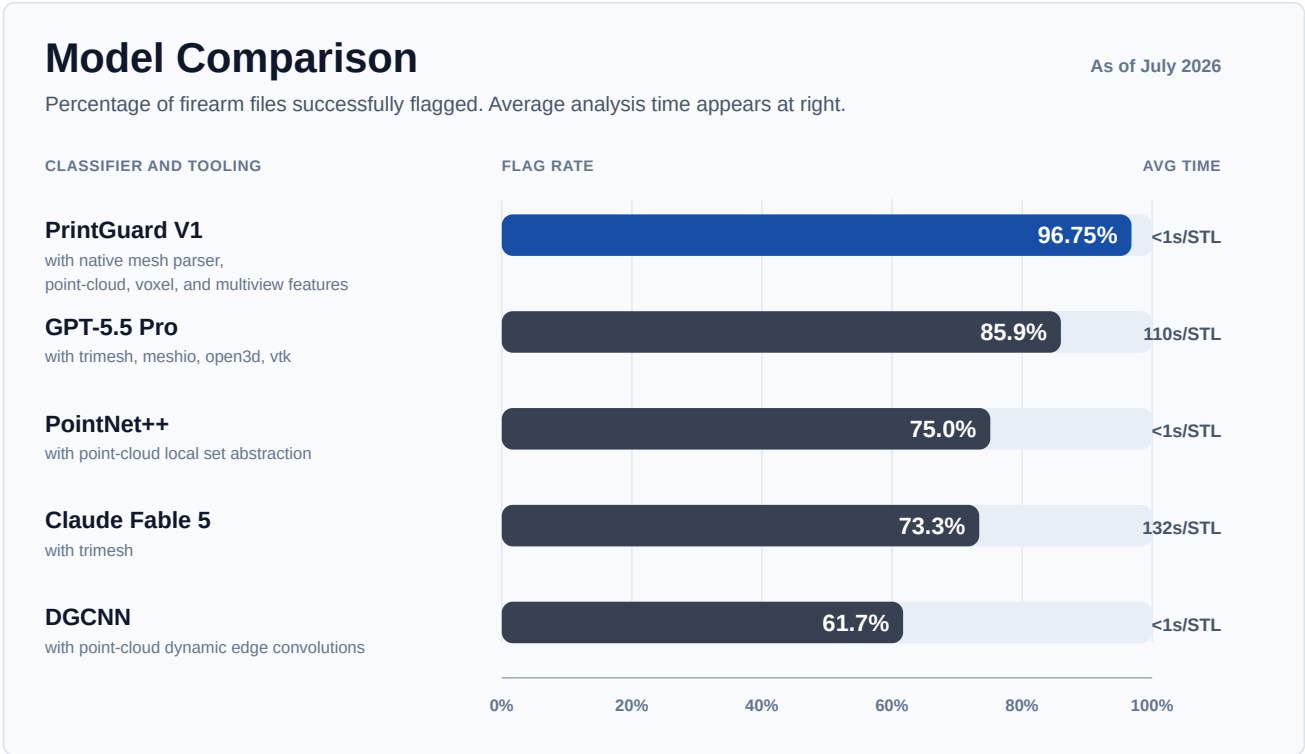
Restricted positive examples sourced through authorized channels; not redistributed.

BENCHMARK SNAPSHOT

The testing set of 10,000 files contained firearms, firearms-specific parts, dual-use parts, near-firearm duplicates, and random benign 3D files.

Detection rate. Percentage of firearm files successfully flagged.

This is the core public-safety metric. Every miss is a firearm-related file that reaches a printer unflagged — and, potentially, an untraceable weapon entering circulation outside the compliance system. California AB 2047 references this directly when it mandates false-negative performance standards for firearm-blueprint detection algorithms.



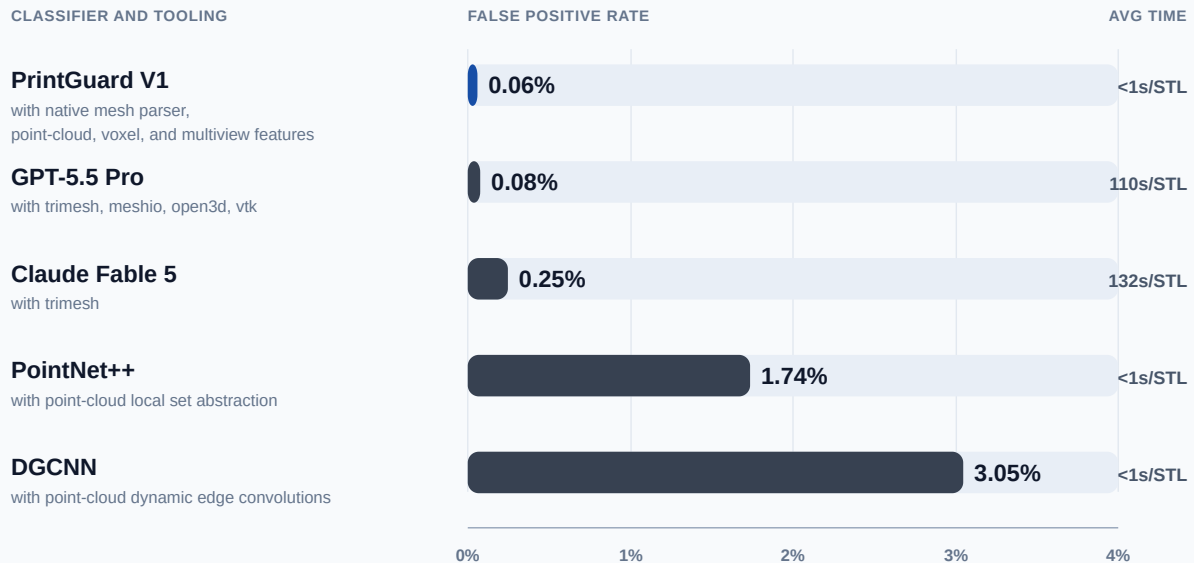
False-positive rate. Percentage of benign files incorrectly flagged. Lower is better.

This is the friction side of the tradeoff. Every benign design wrongly flagged is a legitimate user held up — a classroom project delayed, a prototype stalled, a repair queued for unnecessary human review. AB 2047 requires false-positive standards precisely because a mandate that broadly impedes lawful use will not survive in the field or in court. PrintGuard is engineered to keep this number small so that safety controls do not become a tax on the majority of makers who are doing nothing wrong.

Model Comparison

As of July 2026

False-positive rate on benign files. Lower is better. Average analysis time appears at right.



The benchmark highlights the product thesis: high-capability AI can be valuable for second-pass review but takes considerably longer, and general-purpose 3D classifiers are fast but not tuned for firearm detection. PrintGuard uses a classical machine-learning pipeline for speed while leveraging firearm-specific workflows and datasets to achieve higher accuracy than frontier AI image models.

WHO IT'S FOR

- Printer OEMs that need a credible compliance path.
- Slicer vendors that need pre-print screening without centralizing every file.
- Marketplaces that need pre-publication review of 3D assets.
- Universities, makerspaces, and labs that need intake controls without shutting down open fabrication.
- Print bureaus and enterprise fabrication teams that need job-level screening.
- Public-safety partners that need reproducible file review without broadly distributing restricted CAD data.

THE ASK

We are seeking pilot partners, public-sector sponsors, technical reviewers, and investment support to make PrintGuard a reference implementation for the performance standards contemplated by California AB 2047 and comparable frameworks in New York, Washington, Colorado, and other jurisdictions.

POSITIONING

PrintGuard gives public officials, manufacturers, and institutions a practical middle path: reduce illegal firearm production without imposing blanket surveillance, disabling legitimate tools, or pretending a hash list can solve

a geometry problem. Schools, small manufacturers, researchers, repair shops, and legitimate makers keep the access they need; regulated production keeps the compliance boundary it needs.

The goal is simple: make 3D printing safer without making lawful innovation harder than it needs to be.